

EXPERIMENT - 1

Identification and Observation of Bad Designs

This report documents five examples of bad design observed at VIT University, Vellore. Each entry includes a photograph, description, location, reasons the design is flawed, and a suggested improvement.

Bad Design 1: Exposed Outdoor Electrical Distribution Panel



Description:

An outdoor electrical panel with multiple switches, an isolator, and a Legrand box mounted on a wooden board. Numerous wires hang loose and unsecured from the bottom of the panel, exposed to weather and foot traffic.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.970901, Long 79.164399)

Reasons why the design is bad:

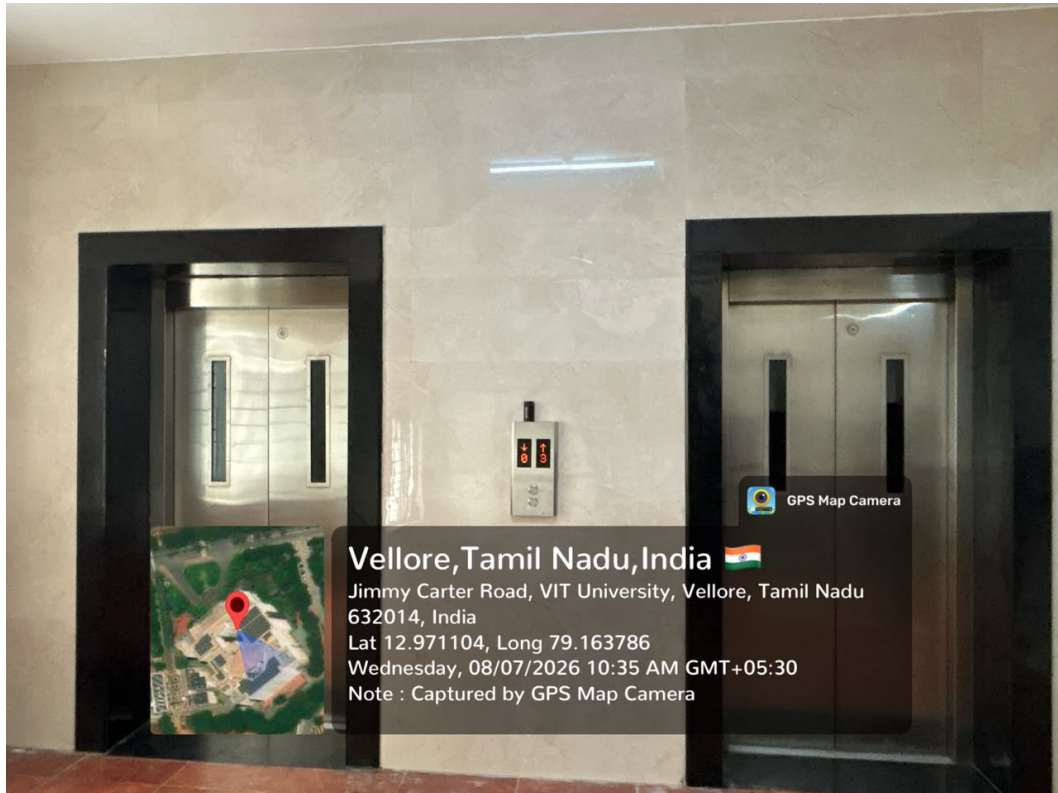
- Exposed, unsecured wiring poses electrocution risk
- No weatherproof enclosure despite outdoor placement

- Loose wires near ground level increase tripping/damage risk
- No visible warning/hazard signage

Suggested Improvement:

- Install a weatherproof (IP-rated) enclosure
- Route and secure wires using conduits or trunking
- Elevate wiring away from ground/foot traffic
- Add hazard warning signage and a restricted-access barrier

Bad Design 2: Elevator Lobby with Shared, Off-Center Indicator Panel



Description:

Two elevators facing each other share a single call/floor indicator panel mounted off to one side, closer to one elevator than the other.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971104, Long 79.163786)

Reasons why the design is bad:

- Panel is not equidistant, making it harder to read from the second elevator
- Only one small display shared between two elevators causes confusion about which car is arriving
- No directional indicator visible from both entrances simultaneously

Suggested Improvement:

- Install separate indicator displays above each elevator door
- Center the shared panel between both elevators if only one is used
- Add audio cues to reduce dependence on visual indicators

Bad Design 3: Unsafe Water Heater Placement in Bathroom



Description:

An electric water heater (geyser) mounted directly above a wash area, with exposed wiring and a switch board on the wall of a wet zone, plus visibly unfinished/peeling wall plaster.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.970394, Long 79.164326)

Reasons why the design is bad:

- Electrical wiring and switches exposed in a high-moisture area is a shock hazard
- Water heater positioned too high for safe/easy access to controls
- Poor wall finishing near electrical fittings increases risk of water seepage into wiring

Suggested Improvement:

- Use moisture-proof, sealed wiring and switch covers rated for wet areas
- Relocate switches outside the wet zone
- Properly seal/finish walls around electrical fittings
- Position heater at an accessible, code-compliant height

Bad Design 4: External Fire-Escape Staircase Without Visible Safety Features



Description:

A tall, narrow metal fire-escape staircase attached to the exterior of a multi-story hostel/academic building, with no visible signage or lighting.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971415, Long 79.164106)

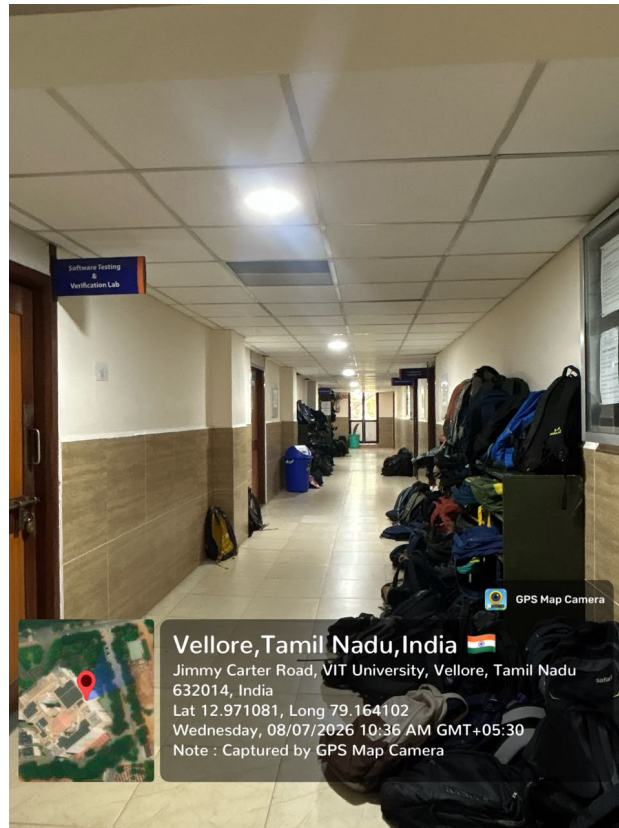
Reasons why the design is bad:

- No visible emergency signage indicating the staircase's purpose
- No visible lighting for nighttime/emergency use
- Narrow structure could bottleneck during mass evacuation
- Overhanging tree branches obstruct part of the structure

Suggested Improvement:

- Add illuminated "Fire Exit" signage at each level
- Install emergency lighting along the staircase
- Trim surrounding vegetation for clear access
- Widen or add a second escape route for large occupancy buildings

Bad Design 5: No Dedicated Storage for Bags Outside Lab



Description:

A corridor outside a lab (Software Testing & Verification Lab) where dozens of backpacks are piled on the floor and stacked on a small improvised stand, since no lockers or storage racks are provided.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971081, Long 79.164102)

Reasons why the design is bad:

- Bags obstruct the walking path, creating a trip/fire hazard
- No designated storage forces students to improvise, creating clutter
- Reduces corridor width available for movement/evacuation
- Unattended bags in open corridor raise security/theft concerns

Suggested Improvement:

- Install proper lockers or a bag rack outside the lab
- Mark a designated bag-storage zone away from the walking path
- Add signage directing students to use provided storage